

Edward Serna

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EDUCATION

B.S., Computer Engineering - University of Texas at San Antonio

December 2025

Relevant Coursework: VLSI Design, Embedded Systems, Electronics, Computer Architecture, Digital Logic Design

Certifications: CompTIA A+, CompTIA IT Fundamentals, Microsoft Technology Associate

TECHNICAL SKILLS

Control Systems: Closed-loop PID design, sensor integration, actuator control, real-time feedback tuning

Programming Languages: C/C++, Python, Verilog, JavaScript, MATLAB, HTML, CSS, Lua

Hardware/ Embedded: MSP430FR5994, ESP32, STM32, FRDM-KL25Z, PIC16F1829, SPI, I2C, UART, GPIO

Tools/Platforms: CMake, Git, PlatformIO, Code Composer Studio, MPLAB X, CLion, VS Code, Linux

Visualization & Web: ImGui, SDL, OpenGL, Next.js, React.js, MongoDB

Languages: English (native), Spanish (conversational)

EXPERIENCE

Research Assistant – ModiFly Team | Unmanned Systems Lab, UTSA

June 2022 – December 2025

- Built the ground control application interface for a team-designed drone ecosystem, developing operator displays, live video, and a virtual simulation environment used for testing and demonstration.
- Contributed across a multi-year platform redesign, from early Python-based waypoint control to a full custom C++ ecosystem with modular ground control, simulation, and drone firmware components, gaining experience coordinating development across hardware and software teams.

Customer Service Associate | Express Fashion

2021-2022

- Developed professional communication, punctuality, and team coordination skills in a high-traffic environment.

PROJECTS

Spotify Playback Display | ESP32, C++, Spotify SDK

Spring 2026

- Designed a touchscreen Spotify remote built on the Elecrow CrowPanel display. Features album art rendering, marquee track titles, playback controls, and live track syncing via the Spotify Web API over Wi-Fi.

PlanterBox IoT Hydroponic System | ESP32, C++, Next.js, MongoDB

Spring/Fall 2025

- Architected an IoT system for plant monitoring and control. ESP32 firmware handles pH, PPM, temperature, humidity, and water level sensors with automated pump/light scheduling. Next.js dashboard hosted on Vercel with real-time charts, plant profiles, and a Wi-Fi setup when device loses connection or first-time setup.

Voice Recognition System | C, MSP430FR5994

Fall 2025

- Integrated a full signal processing pipeline on a resource-constrained microcontroller, implementing structured test procedures and data collection tools to evaluate system accuracy and isolate hardware/software faults.

Autonomous Cup Car - 1st Place, Microcomputer Systems II | ARM Cortex-M0+, C

Spring 2025

- Tuned a closed-loop PID motor controller on an ARM Cortex-M0+ microcontroller, iteratively adjusting control gains and sensor alignment through structured hardware testing to achieve the fastest track time in the class competition.
- Gained hands-on experience with real-time control systems, sensor calibration, gain adjustment, and fault isolation.

Music Glove - 3rd Place, Microcomputer Systems I | PIC16F1829, C

Spring 2023

- Programmed a wearable embedded device that reads flex sensor inputs and maps them to musical outputs across multiple selectable modes, applying interrupt-driven firmware design.